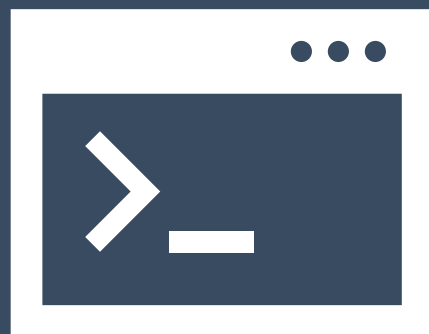


Introduction to Spatial Transcriptomics

<https://tinyurl.com/hbc-spatial>



Harvard Chan Bioinformatics Core

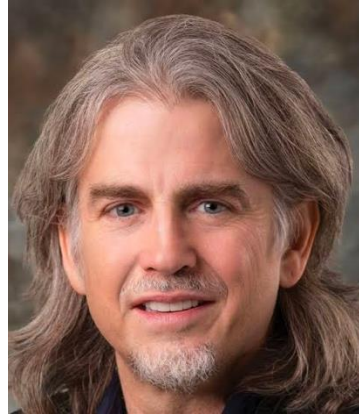


Introductions!





Shannan Ho Sui
Director



John Quackenbush
Faculty Advisor



James Billingsley



Emma Berdan



Elizabeth
Partan



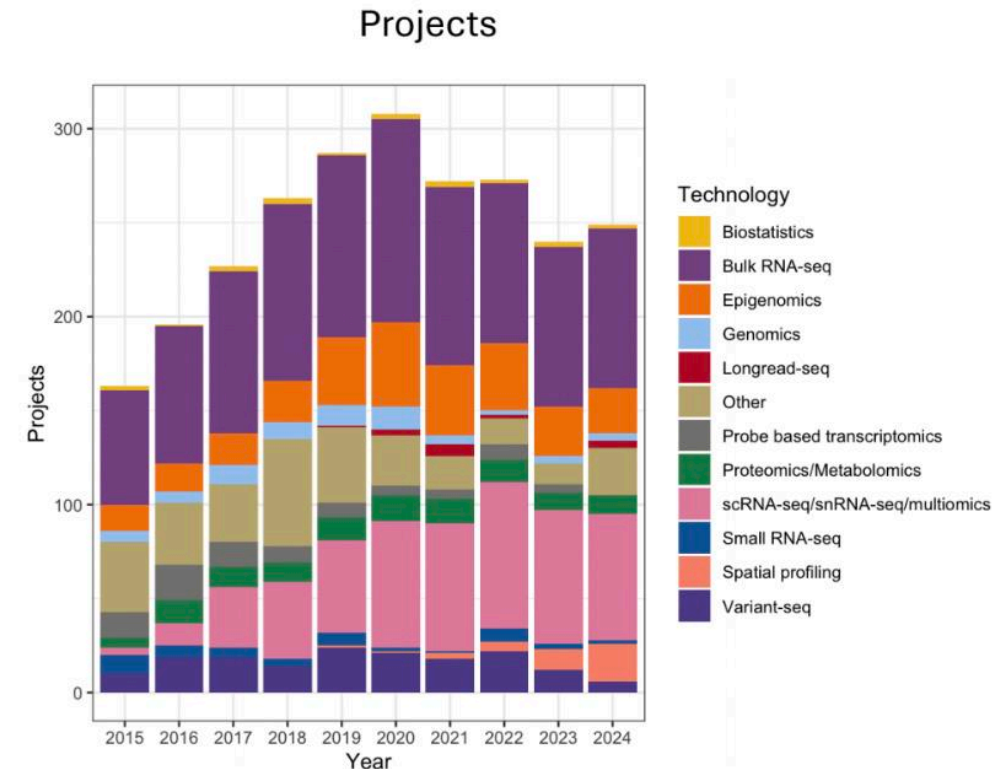
Will Gammerdinger



Noor Sohail

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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HARVARD
T.H. CHAN
SCHOOL OF PUBLIC HEALTH

NIEHS



THE HARVARD CLINICAL
AND TRANSLATIONAL
SCIENCE CENTER



HARVARD
MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Python

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Spatial Transcriptomics

- ❖ Tools for Reproducible Research

- ❖ Current Topics in Bioinformatics

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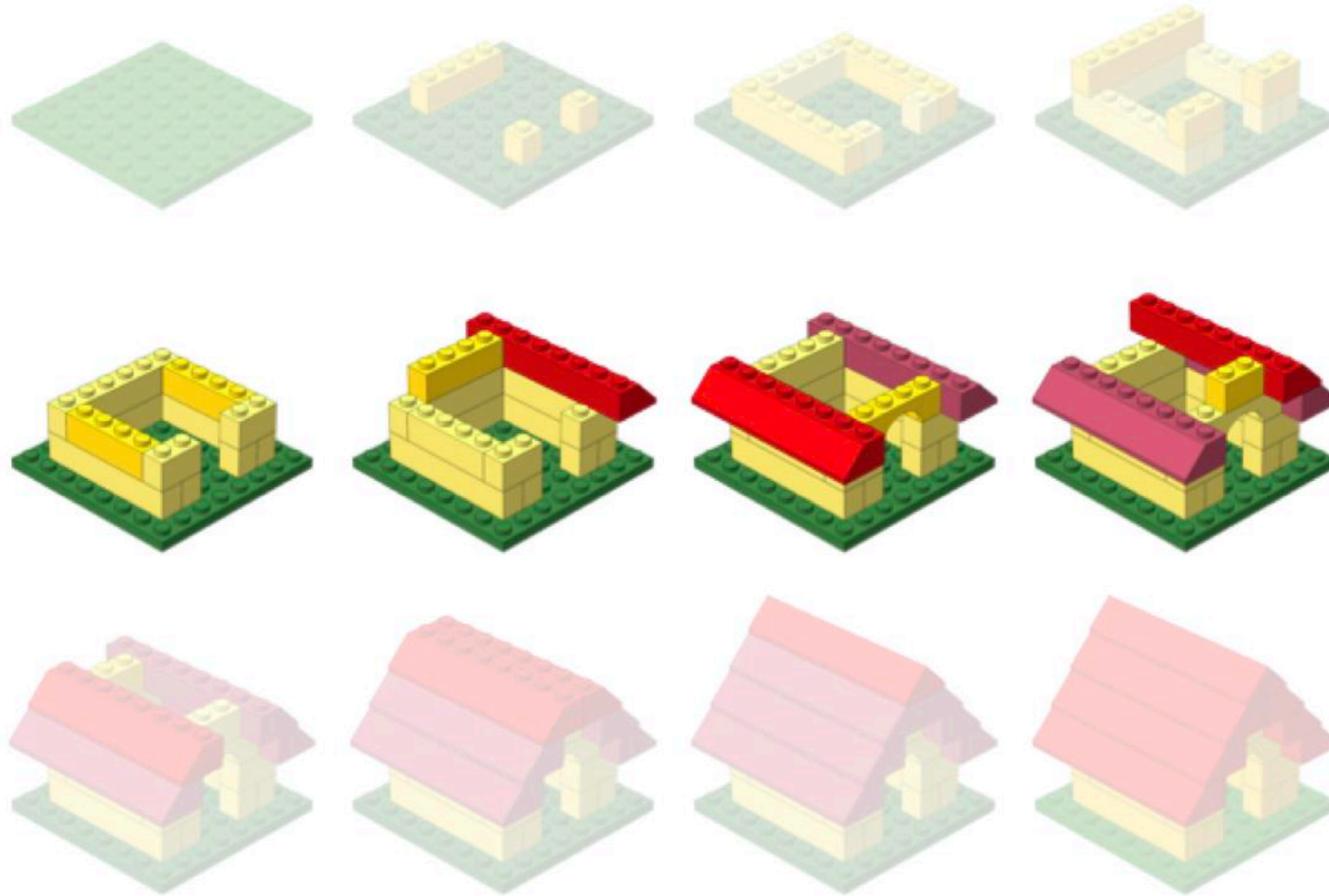
❖ Tools for Reproducible Research

❖ Current Topics in Bioinformatics



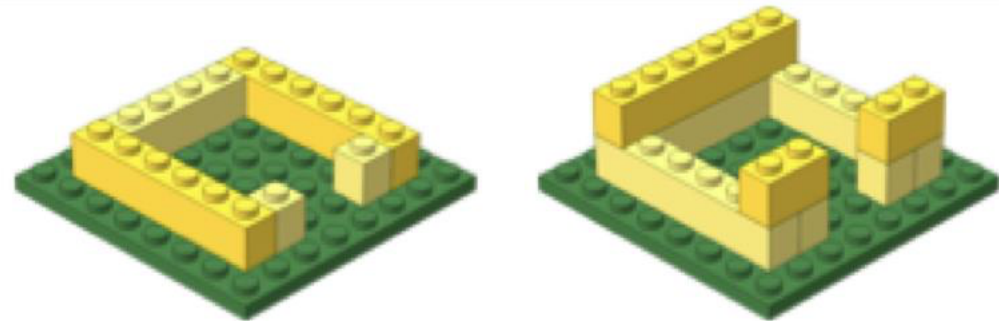
<https://hsph.harvard.edu/research/bioinformatics/training/>

Workshop scope



Bioinformatic Data Analysis

Learning Objectives



- ❖ Experimental design best practices of current spatial transcriptomics technologies
- ❖ Data loading and quality control of Visium HD samples
- ❖ Standard single-cell workflows adapted for spatial data, including highly variable gene selection, PCA, UMAP, kNN, and clustering
- ❖ Cell type annotation with deconvolution using RCTD for sequencing-based technologies
- ❖ Spatial-specific analyses that utilize the physical location of bins/cells on the tissue
 - ❖ Spatial clustering with BANKSY
 - ❖ Spatially variable gene detection with Moran's I
 - ❖ Cell-cell communication with CellChat

Logistics



Course schedule

Spatial Transcriptomics

Schedule

HBC

GitHub

Contact us

Schedule - Spatial Transcriptomics

</> Code

Pre-reading

1. [Spatial Technology Overview](#)
2. [Space Ranger Summary](#)
3. [Download Project](#)

Day 1

Time	Topic	Instructor
10:00 - 10:15	Workshop Introduction	Will
10:45 - 11:45	Introduction to Spatial Transcriptomics	Dr. Pratyusha Bala
10:45 - 10:50	Break	
10:50 - 11:25	Load Visium HD Data	Noor
11:55 - 12:00	Overview of self-learning materials and homework submission	Will

<https://tinyurl.com/hbc-spatial>

Course materials

❖ We continuously update our materials to reflect changes in the field/software

[Spatial Transcriptomics](#) [Schedule](#) [HBC](#) [GitHub](#) [Contact us](#) 

Pre-reading: 

[Spatial Transcriptomics Technologies](#)

Space Ranger Processing

Day 1: 

Day 1 Self-learning: 

Day 2: 

Day 2 Self-learning: 

Day 3: 

Day 3 Self-learning: 

Day 4: 

Spatial Transcriptomics Technologies

[SPATIAL TRANSCRIPTOMICS](#) [EXPERIMENTAL DESIGN](#)

Explore the current landscape of spatial transcriptomics technologies. Focusing on contrasting sequencing-based and imaging-based platforms to show how they differ in resolution, throughput and sensitivity. By the end, you will be informed on which technology is best suited for your biological question.

KEYWORDS
Imaging-based, Sequencing-based, Visium, MERFISH, seqFISH

Approximate time: 30 minutes

Learning objectives

In this lesson, we will:

- Describe the evolution of high-throughput sequencing

On this page

[Learning objectives](#)

[Overview of lesson](#)

[High-throughput sequencing overview](#)

[Sequencing-based technologies](#)

[Imaging-based technologies](#)

[Summary of comparison](#)

[Experimental design considerations](#)

<https://tinyurl.com/hbc-spatial>

Single Screen & 3 Windows

The image is a composite of three overlapping windows, illustrating a 'Single Screen & 3 Windows' setup. The top window is a Zoom meeting interface showing a video call with two participants, a chat window on the right with a message about Python plotting, and a meeting controls bar at the bottom. The bottom-left window is a JupyterLab interface running on a local host, displaying a Python script that reads a CSV file and processes metadata. The bottom-right window is an RStudio interface, showing the R console with version information and a file explorer on the right listing files like 'data', 'figures', 'README.txt', 'results', 'scripts', and 'spatial_transcriptomics.Rproj'.

Our Recommendation

Single Screen & 3 Windows

Zoom

Meeting: Will Gammerdinger's screen

Participants: Noor Sohal, Will Gammerdinger (host)

JupyterLab

localhost:8881/lab/tree/Desktop/ntra_python

File Explorer:

- data
- figures
- results
- data.zip
- ntra_python.py

Code Editor:

```
[28]: metadata = pd.read_csv("data/mouse_exp_design.csv")

[29]: # Get the row names
      metadata.index
      # Get the column names
      metadata.columns

[30]: Index(['sample1', 'sample2', 'sample3', 'sample4', 'sample5', 'sample6',
          'sample7', 'sample8', 'sample9', 'sample10', 'sample11', 'sample12'],
          dtype='object')

[31]: metadata.loc[0, :]

[32]: genotype    WT
      celltype    typek
      replicate    3
      Name: sample1, dtype: object

[33]: metadata.loc[:, 0]

[34]: sample1    WT
      sample2    WT
      sample3    WT
      sample4    KO
      sample5    KO
      sample6    KO
      sample7    WT
      sample8    WT
      sample9    KO
      sample10   KO
      sample11   KO
      sample12   KO
      Name: genotype, dtype: object
```

Zoom Meeting

You to Everyone 8:38 AM

I have a question about how to make figures in Python. How do you make multiple plots at once?

Who can see your messages?

To: Everyone

Type message here...

On this page

- Learning objectives
- Overview of lesson
- High-throughput sequencing overview
- Sequencing-based technologies
- Imaging-based technologies
- Summary of comparison
- Experimental design considerations

Environment is empty

RStudio

Files | Plots | Packages | Help | Viewer | Presentation

Folder | File | Delete | Rename | Settings

g > Training_programs > Workshops > new_workshop_data > spatial_transcriptomics

Name	Size	Modified
..		
data		
figures		
README.txt	1.9 KB	Apr 29, 2026, 8:39 PM
results		
scripts		
spatial_transcriptomics.Rproj	205 B	Apr 30, 2026, 1:12 PM

1:1 (Top Level) | Posit AI: No completions available. | R Script

Console | Terminal | Background Jobs

R - R 4.5.3 · ~/Harvard University Dropbox/HBC Team Folder (1)/Teaching/Training_programs/Wc

R version 4.5.3 (2026-03-11) -- "Reassured Reassurer"

Copyright (C) 2026 The R Foundation for Statistical Computing

Platform: aarch64-apple-darwin20

R is free software and comes with ABSOLUTELY NO WARRANTY.

You are welcome to redistribute it under certain conditions.

Type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

Our Recommendation

Single Screen & 3 Windows

Web Browser

The image displays a Zoom meeting interface with three windows. The top window shows a JupyterLab interface with a file explorer on the left, a code editor in the center, and a terminal at the bottom. The middle window shows a Zoom chat interface with a message from 'You to Everyone' at 8:38 AM. The bottom window shows a file explorer interface with a list of files and folders, including 'data', 'figures', 'README.txt', 'results', 'scripts', and 'spatial_transcriptomics.Rproj'.

Our Recommendation

Single Screen & 3 Windows

Our Recommendation

Single Screen & 3 Windows

The image displays three overlapping software windows, each highlighted with a colored border and a large text label:

- Zoom** (Blue border): A Zoom meeting window titled "Will Gammerdinger's screen". It shows a video call with participants Noor Sohal and Will Gammerdinger. Below the video is a JupyterLab interface with a file explorer on the left and a code editor on the right. The code editor contains Python code for reading and processing CSV data.
- Web Browser** (Green border): A web browser window showing a page titled "omics". The page has a navigation bar with "HBC", "GitHub", and "Contact us". The main content area discusses transcriptomics technologies, focusing on contrasting single-cell RNA-seq and bulk RNA-seq. A sidebar on the right lists "Learning objectives" such as "Overview of lesson", "High-throughput sequencing overview", and "Sequencing-based technologies".
- R Studio** (Purple border): An R Studio window showing the "Console" tab. The console displays the R version (4.5.3) and the path to the R script. The script content is visible, showing the R license and a note about natural language support. The "Files" pane on the right shows a directory structure with files like "figures", "README.txt", "results", "scripts", and "spatial_transcriptomics.Rproj".

Our Recommendation

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete
- ❖ Homework load is heavier near the end of the workshop

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

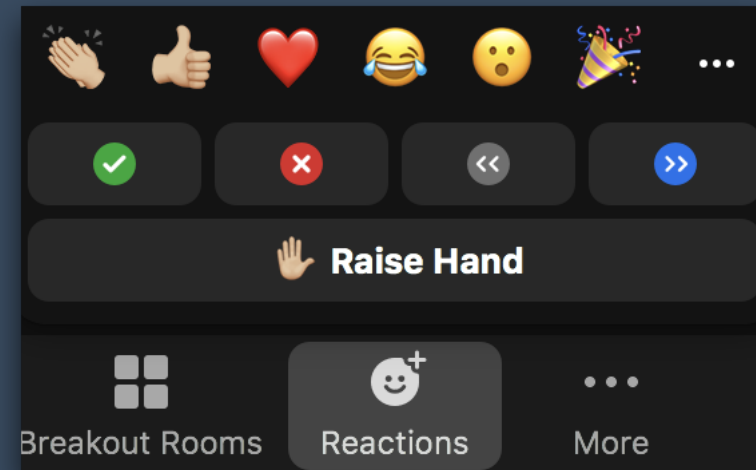
Odds & Ends

❖ Quit/minimize all applications that are not required for class

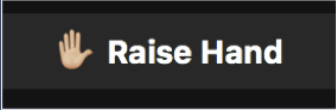
❖ Are you all set?

❖  = "agree", "I'm all set"

❖  = "disagree", "I need help"



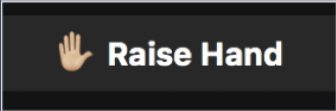
Odds & Ends

- ❖ Questions for the presenter?
 - ❖ Post the question in the Chat window OR
 - ❖  when the presenter asks for questions
 - ❖ Let the Moderator know

Odds & Ends

❖ Questions for the presenter?

- ❖ Post the question in the Chat window OR

- ❖  when the presenter asks for questions

- ❖ Let the Moderator know

❖ Technical difficulties with software?

- ❖ Start a private chat with the Moderator with a description of the problem

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu